# CSCI 5521: Machine Learning Fundamentals (Spring 2022)

## Quiz 4 (Thursday, Apr 21))

**Due on Gradescope at 02:00 PM, Friday, Apr 22**

**Instructions:**

- This quiz has 3+1 questions (1 question for extra credits), 30+2 points, on 2 pages.
- Please **write your name & ID on this cover page**.

1. (**12 points**) Given the data in the figure below, to build a decision tree for classification, what is the attribute you will use to split. Show your work.

| Attribute A | Attribute B | Class Label |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 1 | 0 |
| 1 | 0 | 1 |

2. (**12 points**) Select all correct statement(s) about decision tree and random forest methods:
   (a) Decision tree method is an unsupervised learning method.
   (b) Entropy, Gini Index, and Misclassification Error are all valid options for feature selection in decision tree method.
   (c) Building more decision trees to form a random forest helps alleviate overfitting.
   (d) Decision tree method is more interpretable than random forest method.

3. (**6 points**) List one difference between linear SVM and kernel SVM.

4. (**2 points, extra credits**) Ada is working in a financial firm. Her team is developing algorithms to approve loans and budgets. They are also interested in understanding and analyzing deciding factors in the process. Ada's team has access to a small set of training data with personal information (e.g., age, income, etc.) and their labels. Suggest a machine learning method that they could use. Briefly explain.